Stroke Prediction Classification Project

Final ML Project - Classification | GitHub-ready deliverables

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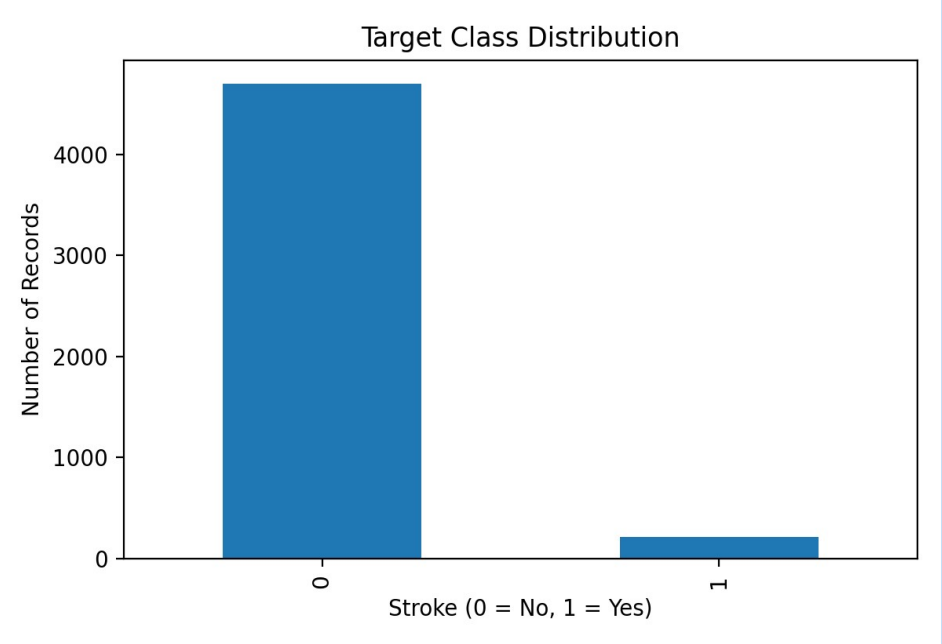


Target class is highly imbalanced: 4,700 no-stroke vs. 209 stroke records.

Data Split and Modeling Workflow

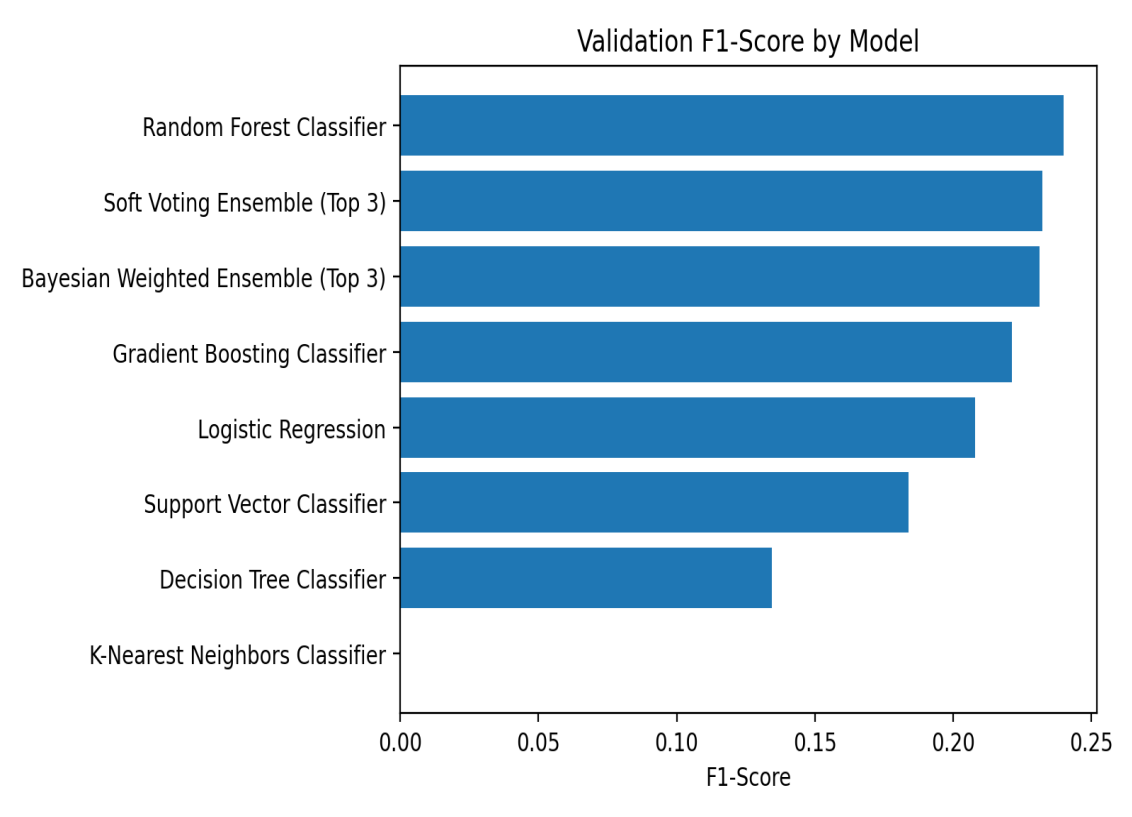
Required Models

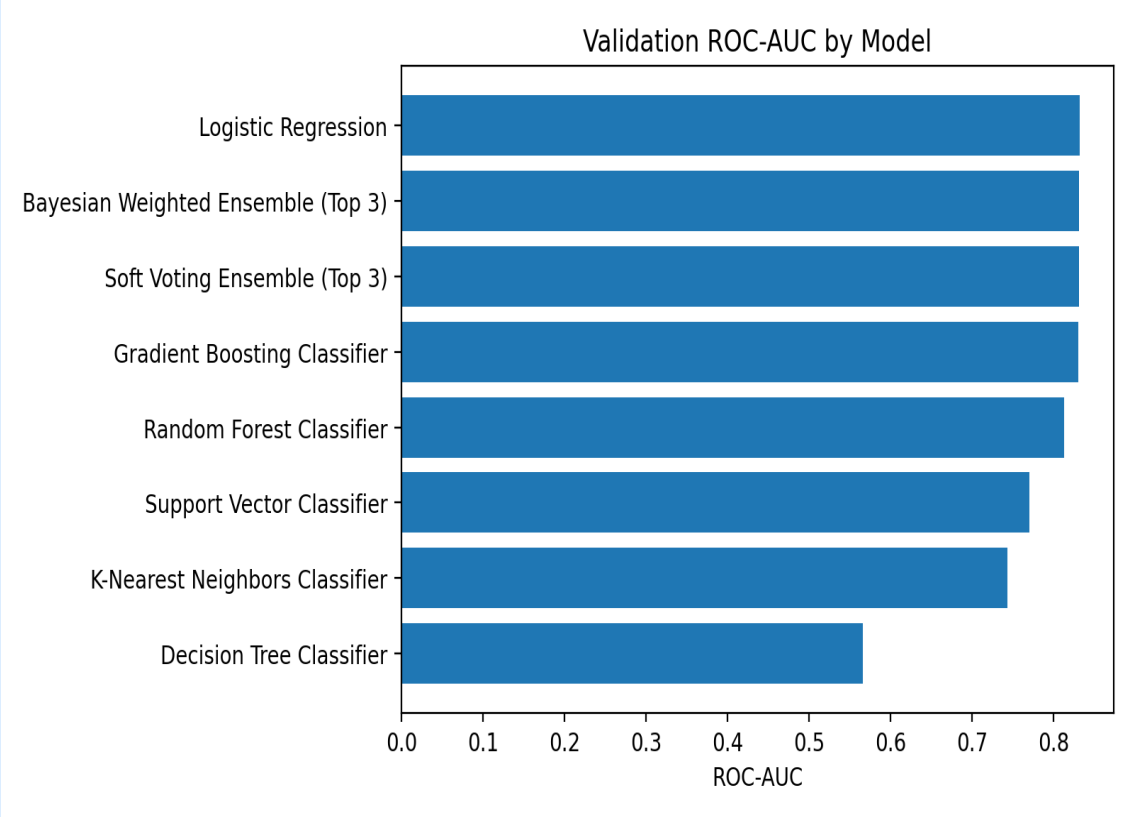
Logistic Regression
Decision Tree Classifier
Random Forest Classifier
Gradient Boosting Classifier
K-Nearest Neighbors Classifier
Support Vector Classifier

Project Setup

Target column: stroke
Split: 70% train, 15% validation, 15% test
Age group one-hot encoded
Numeric features scaled in pipeline
Metrics: Accuracy, Precision, Recall, F1, ROC-AUC

Validation Results: Choose Models Carefully





Accuracy alone can be misleading because the positive stroke class is small. F1 and ROC-AUC give a better comparison.

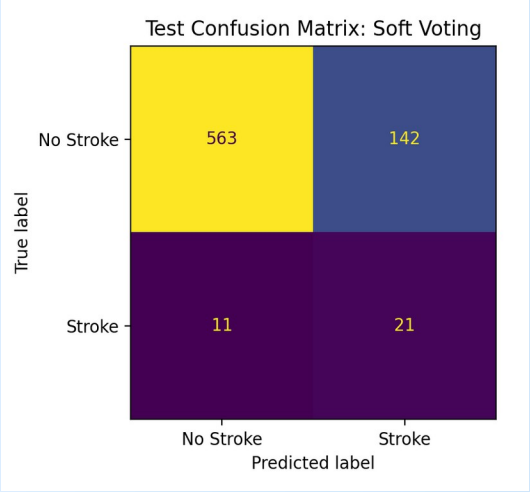
Final Recommendation: Ensemble Model

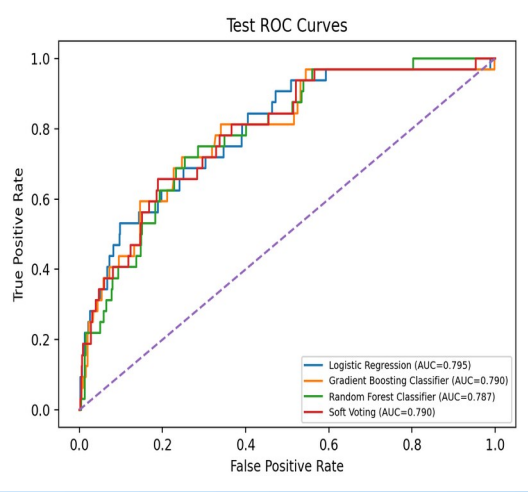
Best Test Result

Soft Voting Ensemble (Top 3)

Accuracy: 0.7924
Precision: 0.1288
Recall: 0.6562
F1-Score: 0.2154
ROC-AUC: 0.7901

The ensemble combined the strongest probability models and produced the strongest F1 balance on the test set.





Conclusion

Use the soft voting or Bayesian weighted ensemble. Future work: tune the probability threshold and test on a larger external healthcare dataset.